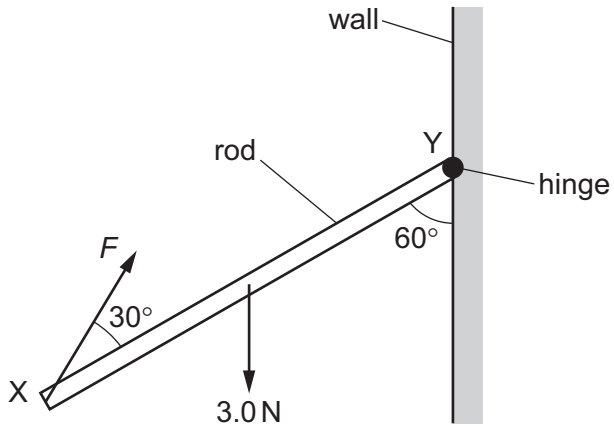
12 The diagram shows a uniform rod, XY, that is freely hinged to a vertical wall at end Y. The rod is at an angle of 60° to the wall.



not to scale

A force *F* acts at an angle of 30° to the rod at end X. The rod has a weight of 3.0 N and is in equilibrium.

What is the magnitude of force *F*?

A 0.87 N **B** 1.5 N **C** 2.6 N **D** 5.2 N